# CardioTwin

## A working cardiovascular digital-twin sandbox

Historical profile + dynamic readings + trained inference

All data and outcomes are simulated. Clinical validity remains unestablished.

Team and college: complete before submission

# The prototype and its boundary

## Focused use case

Explore a five-year hypothetical cardiovascular event probability as a patient twin receives new observations.

## Doctor-facing workflow

Review history, inspect daily changes, compare an input scenario and trace each update.

## Evidence boundary

This is an engineering proof of concept. It does not estimate validated real-patient risk or prescribe treatment.

# The data decision

## Original sample audit

108 Synthea patients. A candidate 2020 index produced only 5 later events, 56 candidate negatives and 7 with insufficient follow-up.

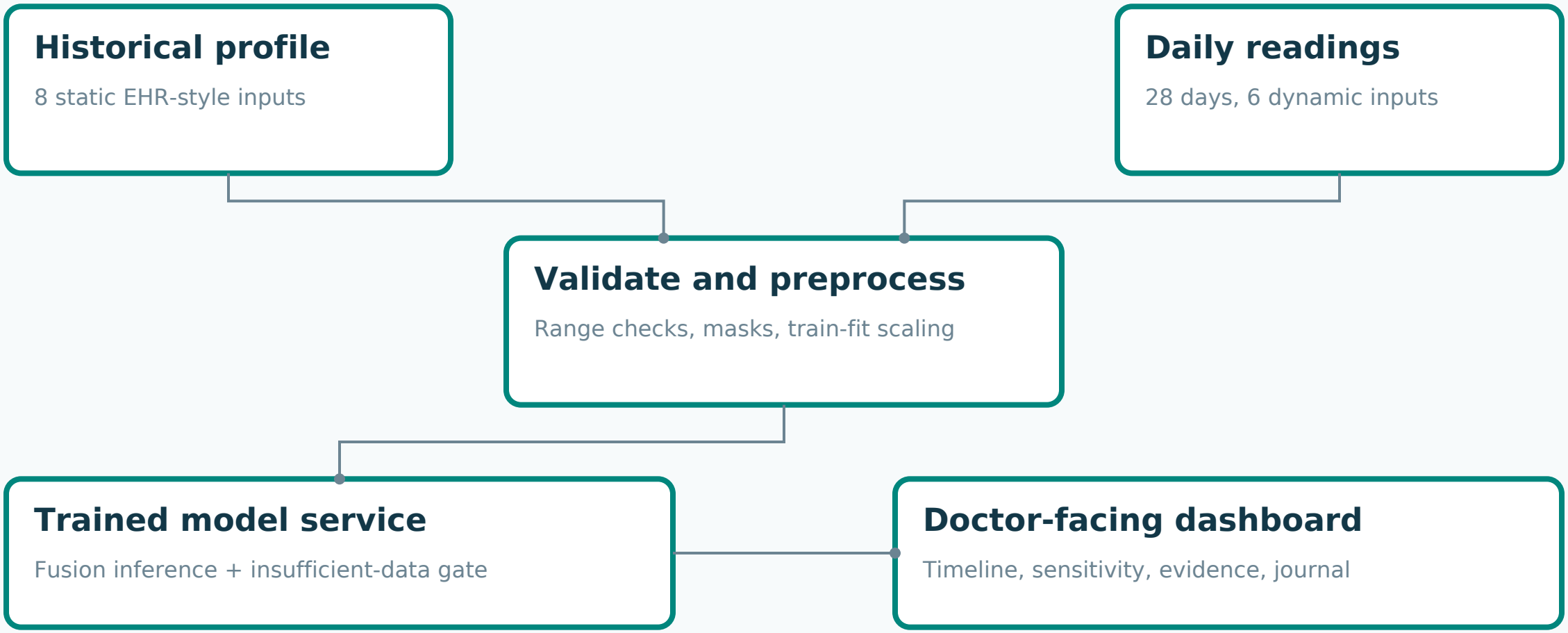
## Matched synthetic benchmark

4,000 new simulated adults, 28 days per patient and a stochastic five-year outcome. No unrelated patient datasets are joined.

## Transparent assumptions

The simulator equation is open in ml/generate.py. Its coefficients are engineering assumptions, not clinical estimates.

# Architecture



# Model comparison protocol

## Four neural models and simple baselines

MLP, 1D CNN, LSTM and GRU fuse a static branch with dynamic inputs. Logistic models test static-only and fused features.

## Patient-level isolation

2,400 training, 800 validation and 800 test patients. Only training patients determine preprocessing statistics.

## Selection before final evaluation

Early stopping and model choice use validation Brier score. The probability threshold stays fixed at 0.5.

# Measured results on simulated patients

| Model | Accuracy | ROC-AUC | Recall | Brier |
|---|---|---|---|---|
| logistic static | 81.38% | 0.814 | 26.6% | 0.133 |
| logistic fusion | 82.12% | 0.840 | 35.6% | 0.123 |
| prevalence baseline | 77.88% | 0.500 | 0.0% | 0.172 |
| mlp | 82.38% | 0.826 | 45.2% | 0.128 |
| cnn | 83.00% | 0.839 | 45.2% | 0.124 |
| lstm | 82.25% | 0.835 | 41.8% | 0.126 |
| gru | 82.50% | 0.838 | 42.9% | 0.125 |

800 held-out patients, 177 events. Logistic fusion wins validation Brier score.

Accuracy alone hides low event recall. These results measure the simulator only.

# The live demonstration

## Update the twin

Ingest a daily summary or replay seven days. The rolling context updates and the backend recomputes the score.

## Handle missing information

Simulate sensor dropout. Insufficient recent data withholds the score rather than presenting a reassuring number.

## Inspect the decision trail

Show input sensitivities, the separate what-if comparison, model evidence and the persisted event journal.

# What makes the prototype credible

## Reproducible results

Generator seed, patient splits, preprocessing, four trained checkpoints and individual test predictions ship with the code.

## Visible limitations

Every screen identifies synthetic data. Scenario changes are model sensitivities, not treatment effects.

## Tested behavior

Automated checks verify split isolation, temporal labels, train-only preprocessing, serving parity and missing-data abstention.

# The route to useful clinical evidence

## Replace the simulator with matched follow-up

Obtain an appropriately governed longitudinal cohort with adjudicated cardiovascular events and dated sensor data.

## Measure transportability

Evaluate calibration, event recall and subgroup performance on external patients, including the intended Indian setting.

## Validate the workflow

Work with clinicians on acceptable missingness, decision thresholds and prospective usefulness before patient-facing deployment.

# Submission and reproducibility

## Included in the repository package

Source, synthetic data, checkpoints, metrics, architecture PDF, presentation PDF, model card and a timed demonstration script.

## Team-owned completion

Add team and college details, record a minimum 20-minute walkthrough, publish the repository and insert the accessible video link.

## Rules source

Unstop: Digital Twin Challenge 2026, Happiest Health. Requirements checked 13 October 2026.